YoungHun Lee

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Profile

Computer Science Ph.D. candidate working on **LLM-based user modeling and personalization**, **RAG**, **RL and LLM post-training**, and **evaluation of subjective judgments**. Industry research experience at Adobe, LG AI Research, and Rakuten. First author on 6 of 8 published papers, including EMNLP 2025.

Education

Aug 2019 to Present — **Purdue University**, West Lafayette, Indiana
Doctor of Philosophy in **Computer Science**, advised by Dr. Dan Goldwasser
- Study Emphasis: **Natural Language Processing** and **Large Language Models**
- Research: Grounding LLMs with personalization and domain knowledge
  - Understanding how individuals judge situations differently [EMNLP 2022, 2025]
  - AI in education; large-scale LLM feedback in Physics class [ACL WS 2026]
  - AI in health; predicting helpfulness of counseling conversation [EACL 2024]
- Award: **Graduate Teaching Award** in 2023.

Sep 2015 to Apr 2017 — **University of Michigan**, Ann Arbor, Michigan
Master of Science in **Information**
- Study Emphases: **Information Retrieval** and **Human-computer Interaction**
- Research: Interaction retrieval and analysis, advised by Dr. Mark W. Newman
- Award: **Outstanding Graduate Student Instructor of the Year** in 2016-2017.

Mar 2008 to Feb 2015 — **Seoul National University**, Seoul, South Korea
Bachelor of Science in **Electrical and Computer Engineering**

Professional Experience

Jun 2024 to Aug 2024 — **Research Intern** at **LG AI Research**, Seoul, Korea
- Developed an AI-safety taxonomy for expert-domain LLMs to guide safe and faithful answers to real-world professional queries
- Crawled and annotated real-world expert Q&A data from StackExchange to ground the taxonomy
- Found novel violation types that general-purpose safety checkers miss

Jun 2023 to Aug 2023 — **Research Scientist/Engineer Intern** at **Adobe**, San Jose, CA
- Proposed *Learning to Reduce*, a small policy language model that removes query-irrelevant rows and columns from long tables so LLMs answer more accurately
- Synthesized row/column relevance training data by executing gold SQL on tables with rows and columns removed; fine-tuned Flan-T5-large with SFT and PPO
- **Improved table QA accuracy by 15%** for longer inputs
- Resulted in an ICML 2024 workshop paper and a U.S. patent application

Jun 2020 to Aug 2020 — **Research Science Intern** at **Rakuten Institute of Technology**, Boston, MA
- Improved state-of-the-art long-tailed product categorization models using unlabeled data augmentation
- Placed top 5 in the SIGIR eCom'20 Data Challenge on multimodal and cross-modal product classification
- Built baseline experiments for contrastive-learning-based long-tailed product categorization, contributing to a weighted contrastive learning method with false-negative control (co-author, ACL 2023 Industry Track)

Selected Research Projects

2024 to 2026 — **Personalized AI Feedback for Students** (with Purdue Physics and Education)

- Built and deployed a real-time feedback system for a physics course with **1,500+ students**, using a self-hosted **70B** open-weight LLM (on-premise)
- Students erring on the targeted quiz question fell from over half in prior semesters to **under 10%**; 78% rated the feedback helpful (*BEA 2026, ACL workshop*)

2022 to Present — **LLMs That Understand How a Specific Person Judges Things** (Ph.D. thesis)

- *SOLAR*: represented each user's history as value trade-offs (e.g., family over money) to retrieve the most relevant past cases, so an LLM predicts that person's judgments more accurately
- **79.9 vs. 48.9 macro-F1** against the best per-user fine-tuned baseline (DeBERTa-v3, RoBERTa, BART, FLAN-T5) on Reddit (EMNLP 2025)
- *(under review)* Extended to counterfactual RAG that avoids spurious matches and contextual dissonance in a user's history

Skills

Programming	**Python**, C++, SQL; **PyTorch**, TensorFlow, **Hugging Face** (Transformers, PEFT, TRL), FlashAttention, vLLM
Training & Inference	SFT, **RLHF (PPO, DPO)**, reward modeling, LoRA/QLoRA, quantization, Inference scaling, LLM applications, AI safety
Retrieval & Modeling	**RAG**, dense retrieval, reranking, FAISS, **user modeling**, contrastive learning, representation learning, explainable AI
Evaluation & Infra	LLM-as-a-judge, human evaluation, A/B testing; Linux, Git, Docker, conda, W&B, GPU training and inference on SLURM clusters

Publications

Jul 2026 — **Younghun Lee**, Amir Bralin, Nobel Sanjay Rebello and Dan Goldwasser, *Toward Just-in-Time Adaptive Feedback: Enhancing Student Learning via Knowledge-Grounded LLM,* ACL 2026 Workshop (BEA 2026), USA

Nov 2025 — **Younghun Lee** and Dan Goldwasser, *SOLAR: Towards Characterizing Subjectivity of Individuals through Modeling Value Conflicts and Trade-offs,* EMNLP 2025, China

Jul 2024 — **Younghun Lee**, Sungchul Kim, Ryan Rossi, Tong Yu, Xiang Chen, *Learning to Reduce: Towards Improving Performance of Large Language Models on Structured Data,* ICML 2024 Workshop on Long-Context Foundation Models

Mar 2024 — **Younghun Lee**, Dan Goldwasser, Laura Schwab-Reese, *Towards Understanding Counseling Conversations: Domain Knowledge and Large Language Models,* Findings of EACL 2024, Malta

Jul 2023 — Tianqi Wang, Lei Chen, Xiaodan Zhu, **Younghun Lee**, Jing Gao, *Weighted Contrastive Learning With False Negative Control to Help Long-tailed Product Classification,* ACL 2023 Industry Track, Canada

Dec 2022 — **Younghun Lee** and Dan Goldwasser, *Towards Explaining Subjective Ground of Individuals on Social Media*, Findings of EMNLP 2022, UAE

Jul 2020 — Hou Wei Chou, **Younghun Lee**, Lei Chen, Yandi Xia, Wei-Te Chen, *CamemBERT and BiT Feature Extraction for Multimodal Product Classification and Retrieval*, SIGIR eCom, China

Oct 2018 — **Younghun Lee**, Seunghyun Yoon and Kyomin Jung, *Comparative Studies of Detecting Abusive Language on Twitter*, EMNLP 2018 Workshop on Abusive Language Online, Belgium